Yinan Bu

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EDUCATION

University of California, Berkeley — Berkeley, USA
Ph.D. in **Industrial Engineering and Operations Research** — Sep 2026 – Present
Advisor: Prof. **Xin Guo**

University of Science and Technology of China (USTC) — Hefei, China
B.Sc. in **Statistics**, School of the Gifted Young — Sep 2022 – Jul 2026
GPA: 4.02/4.30 (92.33/100) Major GPA: 4.12/4.30 (93.63/100)

Research Interests: Reinforcement Learning, Diffusion Models, Statistical Machine Learning

Skills: C, Python (PyTorch), R (Rcpp), LaTeX, Mathematica, MATLAB

PUBLICATIONS & MANUSCRIPTS

- F. Jiang, **Y. Bu**, S. Wu, G. Xu, J. Zhu. (2026). *Efficient Synthetic Network Generation via Latent Embedding Reconstruction*. ICML 2026, PMLR 306:52043–52090.
- D. Wu*, **Y. Bu***, X. Wang. *Hard Aggregation, Prototype Adaptation, and Safe Selection in Hyperdimensional Computing.* Submitted to the Journal of the American Statistical Association (JASA).

** Equal contribution.*

RESEARCH EXPERIENCE

Efficient Synthetic Network Generation via Latent Embedding Reconstruction — Jul 2025 – Dec 2025
Advisor: Prof. **Gongjun Xu**, Prof. **Ji Zhu**

- Developed **SyNGLER**, a general, efficient framework for generating synthetic networks by leveraging latent space network models with a distribution-free generator over learned latent embeddings. Extended this framework to attributed networks and proposed **SyNGLER-Attr**.
- Built scalable pipelines for a diffusion-based latent embedding generator and a bootstrap-based latent embedding resampler, preserving key network characteristics while enabling efficient training and sampling with lower computational cost than many existing deep architectures (GitHub).
- Conducted empirical studies on both simulated datasets and real-world datasets, showing that the proposed method efficiently generates networks that more faithfully preserve key characteristics than existing approaches.

Hard Aggregation, Prototype Adaptation, and Safe Selection in Hyperdimensional Computing — Apr 2024 – Jan 2026
Advisor: Prof. **Xueqin Wang**

- Established a sign-kernel limit for HDC classifiers; quantified the $\pi/2$ effective-dimension penalty of one-bit quantization and proved context-dependent signal cancellation renders class-mean prototypes uninformative.
- Developed **Unified Gated-Prototype HDC** with integer-weighted feature aggregation and context-routed local prototypes; derived theoretical guarantees for feature weighting, context recovery, and classification risk.
- Established **finite-sample no-harm guarantees** for model selection; reduced simulated classification error from **50.0% to 15.5%**, with routed HDC outperforming other hard HDC methods on WISDM.

Efficient Graph Trend Filtering via Differential Operator Grouping — Nov 2024 – Feb 2025
Advisor: Prof. **Xueqin Wang**

- Worked on graph trend filtering (minimizing ℓ_1 norm of discrete graph differences) to recover piecewise-smooth signals; examined the ADMM trade-off between convergence speed and subproblem solvability.
- Proposed **D**ifferential **O**perator **G**rouping–bas**e**d ADMM (**Doge-ADMM**), grouping differential operators to get closed-form subproblems and parallel updates.
- Built a parallel implementation for first- and second-order cases and achieved up to **30**$\times$ speedup compared with standard ADMM solvers (GitHub).

ACADEMIC PROJECTS

Analysis of the Government Pension Fund of Norway Jun 2025 – Jul 2025
Supervisor: Prof. Canhong Wen (Department of Statistics and Finance, USTC)

- Independently designed, implemented, and deployed an RShiny website for the Norwegian Government Pension Fund Global (NBIM) with interactive Plotly charts and a Leaflet world map (RShiny).
- Conducted a comprehensive analysis that integrated statistical summaries with mapped visualizations and annotated trends, including embedded figures and map snapshots (GitHub).

Uncertainty-Aware Time-Series Forecasting via Conformal Prediction Dec 2024 – Jan 2025
Supervisor: Prof. Yu Chen (Department of Statistics and Finance, USTC)

- Reproduced the conformal prediction framework in Stankeviciute et al. (2021) for probabilistic time-series forecasting, implementing model-agnostic, distribution-free prediction intervals.
- Conducted experiments on simulated and real-world datasets (AR/ARIMA, sales, air quality, COVID-19), showing robust uncertainty quantification and competitive interval performance versus standard baselines.

SELECTED UNDERGRADUATE COURSEWORK

Mathematics: Real Analysis, Functional Analysis, Mathematical Analysis, Linear Algebra, Differential Equations.
Probability & Statistics: Probability, Mathematical Statistics, Applied Stochastic Processes, Regression Analysis, Multivariate Analysis, Nonparametric Statistics.
Machine Learning & Computing: Machine Learning, Statistical Algorithms, Time Series, C Programming.

AWARDS & HONORS

Xingjian Scholarship (alumni-funded scholarship recognizing excellence in athletics) 2026
Guo Moruo Scholarship Nomination (most prestigious scholarship at USTC) 2025
China National Scholarship (highest scholarship from Ministry of Education of China) 2025
Yang Ya Alumni Fund Scholarship (top 5 female students in School of the Gifted Young) 2024
Excellent Student Scholarship (top 10% students in School of the Gifted Young) 2023

ADDITIONAL INFORMATION

Teaching Assistant:

- C Programming Language (Undergraduate), *Instructor: Prof. Lixiang Tan* Sep 2024 – Feb 2025
- Linear Algebra I (Undergraduate), *Instructor: Prof. Junchao Shentu* Mar 2025 – Jun 2025

Code & Software:

- **SyNGLER**: A synthetic network generation framework based on latent embedding reconstruction (GitHub).
- **Doge-ADMM**: Differential Operator Grouping–based ADMM (GitHub).
- **NBIM**: Interactive visualization of the Norwegian Government Pension Fund global holdings (GitHub, RShiny).

Leadership & Activities:

- Member, School of the Gifted Young Football Team, USTC 2022 - 2025
- Three Gold Medals, USTC University-Wide Track and Field Championships 2023 - 2025
- Flutist, USTC Chinese Orchestra 2022 - 2025